Standardization and Normalization (Feature Scaling)

standardization and normalization are preprocessing techniques that ensure data is prepared in a way that it

1) maximizes the effectiveness of machine learning algorithms,
2) improves model performance, and
3) aids in the interpretability of results.

Their application depends on the specific characteristics of the data and the requirements of the modeling techniques being used.

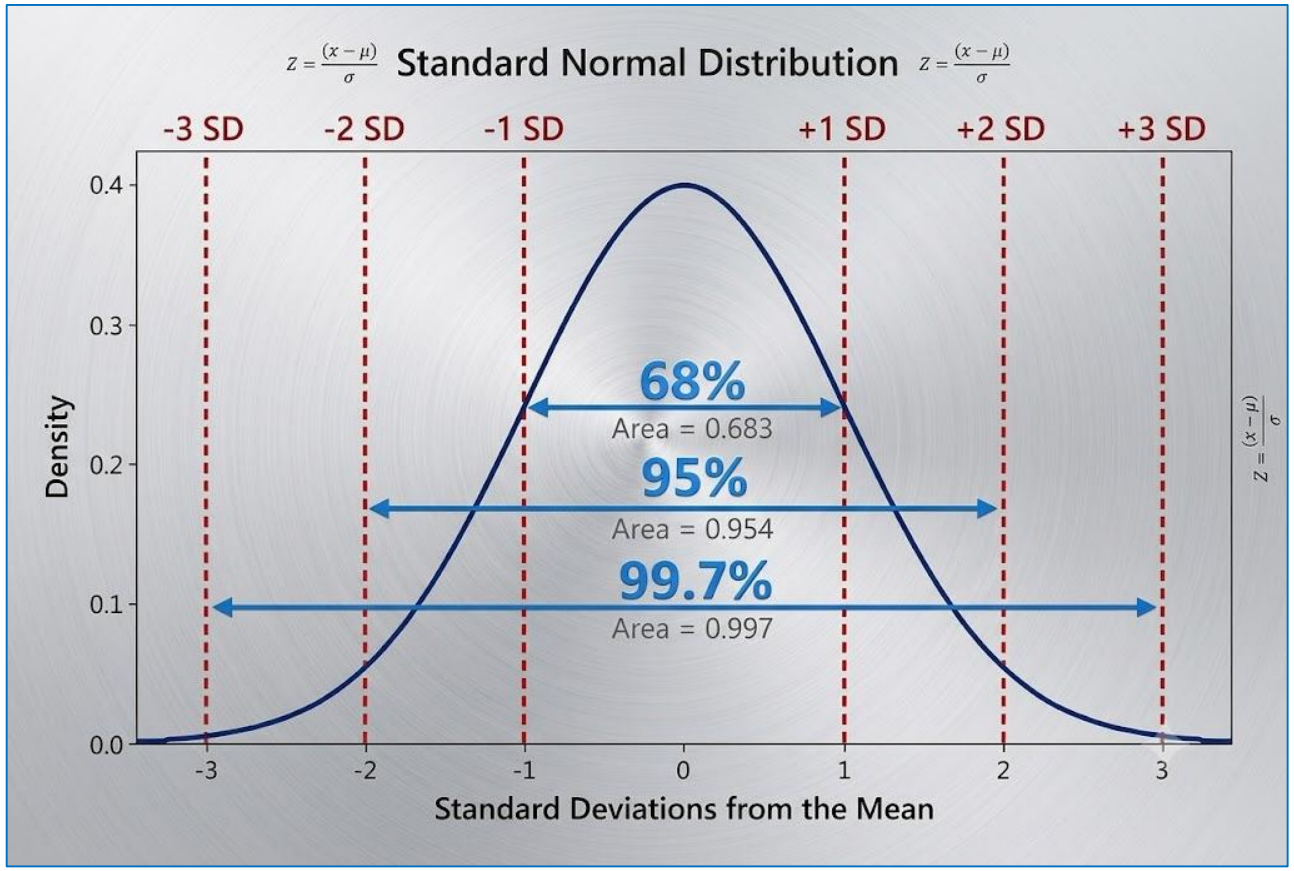
How the Empirical Rule Maps to Z-Scores

When your data is converted to the Standard Normal Distribution (where mean = 0 and standard deviation = 1), the rule breaks down exactly by the Z-score numbers:

68% of the data falls between a Z-score of **-1 and +1.**

95% of the data falls between a Z-score of **-2 and +2.**

99.7% of the data falls between a Z-score of **-3 and +3.**



Standardization (Standard Normal Distribution:)

Converting a **normal distribution** in to **standard normal distribution** is called Standardization. The standard normal distribution is also called the z-distribution; **any normal distribution can be standardized by converting its values into z-scores formula.**

Where mean become zero and standard deviation becomes one

Calculate the Standard Deviation:

X [1,2,3,4,5]

$$\text{Z Score} = \frac{x_i - \mu}{\sigma}$$

$\mu = \text{Mean}$

σ= Standard Deviation

1	$\frac{1-3}{1.41} =$	-1.414
2	$\frac{2-3}{1.41} =$	-0.707
3	$\frac{3-3}{1.41} =$	0
4	$\frac{4-3}{1.41} =$	0.707
5	$\frac{5-3}{1.41} =$	1.414

Using the standard normal distribution, or more specifically, standardizing data to have a mean of 0 and a standard deviation of 1, offers several advantages in statistics and data analysis.

Here are some of the main reasons why the standard normal deviation is commonly used:

- **Comparability:**
- **Simplification:**
- **Outlier Detection:**
- **Feature Scaling:**

Normalization (Min Max Scaler)

Normalization is the process of organizing the data in the database. · Normalization is used to minimize the redundancy from a relation or set of relations.

It restricts the data from 0 to 1

Normalize the below data

X =[1,2,3,4,5,6,7,8,9,10]

$$y = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Values	Y
1-1/10-1	0.000
2-1/10-1	0.111
3-1/10-1	0.222
4-1/10-1	0.333
5-1/10-1	0.444
6-1/10-1	0.556
7-1/10-1	0.667
8-1/10-1	0.778
9-1/10-1	0.889
10-1/10-1	1.000

$$y = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$\text{Z Score} = \frac{x_i - \mu}{\sigma}$$

Normalization	Standardization
The minimum and maximum values of features are used for scaling	Mean and the standard deviation is used for scaling.
It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
It is really affected by outliers.	It is much less affected by outliers.
Scikit-Learn provides a transformer called MinMaxScaler for Normalization.	Scikit-Learn provides a transformer called StandardScaler for standardization.
This transformation squishes the n-dimensional data into an n-dimensional unit hypercube.	It translates the data to the mean vector of the original data to the origin and squishes or expands.
It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.
It is often called Scaling Normalization	It is often called Z-Score Normalization.